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Returns Policy and Quality Risk in E-Business

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In this article, we investigate the interplay between returns policy, pricing strategy, and quality risk. We define quality risk as the possibility of product misfit, defect, or unconformity with the consumers' perception. These notions of quality risks differ in return policy restriction, residual values, and whether it is possible to unambiguously reduce the probability of mismatch. Using a stylized two-segment market setting, we demonstrate that consumer returns are offered only when the high-segment consumers incur a higher hassle cost, and both the quality risk and the valuation of the low segment are moderate. Moreover, it is possible to wisely design the returns policy that eliminates all inappropriate returns. Furthermore, the seller with a high-quality risk may offer a refund that exceeds the selling price, which provides a theoretical ground and specific operating regime for the satisfaction guaranteed policy used in some e-tailers. In contrast, when the quality risk is relatively low, further improvement on mitigating the quality risk may not necessarily benefit the seller. Finally, we observe that the restocking fee may be non-monotonic in product quality; thus, a more generous returns policy does not necessarily indicate a lower quality risk.

Key words: consumer returns; quality risk; restocking fees; satisfaction guaranteed
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1. Introduction

Electronic business has dramatically changed consumers' purchasing behaviors. As an example, if one intends to purchase a pair of shoes, she can simply browse through various websites, such as Amazon, Famous Footwear, OnlineShoes, Payless, RoadRunner Sports, Shoebacca, ShoeBuy, ShoeMall, and Shoes.com. Essentially, in this "clicks-and-mortar" model, a complete transaction requires simple clicks online and the consumer can then wait at home for the product delivery. From the consumers' viewpoint, however, this e-business revolution gives rise to an additional concern. Compared with the traditional "bricks-and-mortar" supply chains, consumers who intend to engage in Internet transactions now are unable to actually see and inspect the products before making their purchases. In response to this significant uncertainty about product quality, Internet manufacturers/retailers typically offer returns policies to the consumers as a protection. Roger and Tibben-Lemke (1999) conducted a survey and found that among the respondents, 63% believe that one of the most important tools for Internet manufacturers/retailers to stay competitive is to offer clear and attractive returns policies. In two other independent surveys (Pinkerton 1997, Trager 2000), more than 70% of consumers claim that they typically evaluate returns policies of the stores before they decide to shop.

As we browse the Internet, the returns policies in e-business vary across industries and stores. Along with the conventional 100% money-back guarantee, there are other surcharges, restrictions, time window limits, and special instructions on labeling, packaging, and tagging. Some Internet manufacturers/retailers charge handling/restocking fees (typically ranging from 15% to 50%) that depend on the condition of the returned products (whether the products are opened, used, marked, etc.), and the shipping costs of the original shipment or the returns are not refundable. Others are rather generous in that the return shipping costs are covered either when the return results from consumer satisfaction (essentially no-questions-asked) or a manufacturing defect. Furthermore, some e-tailers charge restocking fees and at the same time cover consumers' shipping costs of returns (in case of manufacturer defect or e-tailer's error). From the 13 Internet stores that sell shoes and their contemporary returns policies listed in Table 1, there seems to be a huge variation on the returns policies among Internet stores, even if the merchandise is similar. Of particular interest is the "free returns in-store" policy (also adopted by some leading retailers with large retail chains, such as Target). Essentially, these Internet

Table 1 Various Returns Policies of Internet Shoes Stores

Internet stores	Shipping rates	Return policies
Amazon.com	Free shipping on 25+	Standard return shipping rates deducted
Endless.com	Free shipping	Free return shipping
Famous Footwear	$4.50 shipping	Free returns in-store
FinishLine	$7.99 shipping	Free returns in-store
OnlineShoes.com	Free shipping	Free exchanges
Payless	$5.95 shipping	Free returns in-store
PiperLime	Free shipping	Free return shipping
RoadRunner Sports	$7.99 shipping	Free returns in-store for online purchases
Shoebacca.com	Free shipping	Free return shipping
ShoeBuy.com	Free shipping	Free return shipping
ShoeMall.com	Free shipping	$6.95 deducted from refund
Shoes.com	Free shipping	Free return shipping
Zappos.com	Free shipping	Free return shipping

stores allow the consumers who purchase online to return the products either to their Internet center or to a physical store, whereas sending back to their Internet center requires either additional shipping rates or refund deductions. This variation is also prevalent in other apparel/home appliance e-retailers, such as KayakProShop, Akinaiblog Japan Shop, The Yellow Tags, Soccer Shirts Online, and Massive24.com.

This article offers an economic rationale for why various Internet sellers may adopt different returns policies. To fix ideas, we call the Internet seller the "manufacturer" throughout this article. We show that, partially in contrast with the conventional wisdom, the product quality is not necessarily positively related to the generosity of the associated returns policy. We define product quality broadly as "the performance, features, styling, and other product attributes that enhance the fitness for use of utility for the consumers" (Fine 1986; see also Garvin 1984). Consequently, product misfit (Su 2009, Swinney 2011), product defect (Emons 1989, Grossman 1981, Shieh 1996), and product unconformity to the consumers' perception ("merchandise didn't match description" in Moorthy and Srinivasan 1995) can all be accommodated in our unified framework. Note that these notions may involve some subjective assessments or perceptions by the consumers because the consumers are not required to provide any evidence for the legitimate reason of returns. Furthermore, the flexible framework we construct in this article also allows us to investigate the following research questions. When should the manufacturer allow returns? If the returns are allowed, when is the restocking fee needed, and when should the manufacturer offer the hassle-free returns policy (by covering the hassle costs)? Is it necessarily true that the manufacturer that sells a higher quality provides a more generous returns policy? How does the returns policy affect the incentive of quality design, channel profit, and consumer welfare? Is the manufacturer always better off by improving the product quality?

In our stylized model, a monopolistic manufacturer intends to sell his products to consumers with heterogeneous preferences. The products may either meet the consumers' expectation (i.e., function well or match the consumers' need) or inappropriate (i.e., breakdowns occur, merchandise does not match the descriptions), which cannot be detected by the consumers at the time of purchase. There are two consumer segments. Consumers within each segment are homogeneous, and a low-segment consumer obtains a lower valuation from using the perfectly functioning product than a high-segment consumer does. As consumers may be hesitant to purchase the product, the manufacturer intends to offer returns as a protection. In addition to the valuation heterogeneity, the consumers are also heterogeneous in their hassle costs upon returning the products. In our basic framework, high-segment consumers incur a higher hassle cost for returning the products, as in the online world the hassle costs primarily arise from the consumers' time values. We use "inappropriate products" to denote the products that are beyond the consumers' expectation, and focus on the situation in which the returns are inefficient from the society's perspective, that is, the consumers' hassle cost is unambiguously higher than the manufacturer's salvage value. The quality is exogenously given, and we allow the refunds to be zero, partial, full, or to even exceed the original selling price.

Before we discuss our findings, it is helpful to elaborate on the subtle differences among the three notions of quality risks—product defects, product misfit, and product unconformity. Specifically, in the context of product defects, an inappropriate product generates no utility for the consumers, the manufacturer is regulated to offer (at least) a full return, and may invest in quality improvement to unambiguously reduce the chance of product inappropriateness. In the case of product misfit, the consumers may obtain some residual value from using an inappropriate product; on the manufacturer's side, he is not bound to offer full returns, but he is unable to unambiguously reduce the probability of mismatch because of the intrinsic heterogeneity of consumer tastes. The notion of product unconformity lies in between these two polar cases: inappropriate products generate residual values, the manufacturer is not bound to offer full returns, and quality improvement can be implemented to increase the probability of conformity. As these notions differ in three dimensions—the return policies, the residual values, and the quality improvement, our discussions are centered around

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these intrinsic characteristics instead of pointing directly to any specific notion of quality risk.

We start with the scenario in which the high-segment consumers incur a higher hassle cost. In this scenario, we show that the manufacturer uses returns to implicitly distinguish the consumer segment through their self-selection behaviors even if returns are socially inefficient. Furthermore, although the manufacturer is unable to inspect the product inappropriateness, he can design the refund wisely so that products that function well never get back to the store. This essentially shows that a careful design of returns policy may eliminate the "inappropriate returns," which historically has been a primary concern for numerous retailers on a variety of products (Hess et al. 1996). In addition, consumer returns are offered only when the product quality is moderate and the valuation of the low segment is moderate.

We next examine how the product quality affects the manufacturer's pricing strategy and the refund amount. We find that when the product quality is relatively low, the manufacturer may *over-compensate* the consumers for returning the product (i.e., the refund may exceed the selling price), as the manufacturer is unable to set a high selling price for a low quality product but he still has to compensate the consumers for returning the inappropriate products. This strategy corresponds to the "satisfaction guaranteed" concept proposed by Fruchter and Gerstner (1999) and speculated in Moorthy and Srinivasan (1995). When the product quality is relatively high, further improvement on product quality may *not* induce a higher selling price (and the corresponding refund). Our results thus identify an intrinsic rationale for the manufacturer's aversion to technological improvement or information provision in the context of consumer returns. The restocking fee (measured in the percentage of selling price) may be *non-monotonic* in product quality, as the manufacturer with a relatively high-quality risk overcompensates the consumers' returns and the refund flattens out when the quality risk is low. It is worth mentioning that in our context, the absolute amount of refund and selling price are both increasing in the product quality, thereby leading to a positive correlation between the refund and price. This result provides a theoretical justification of the empirical finding in Ancarani et al. (2009).

When the quality of the product is the manufacturer's discretion, we provide complete comparative statics of how several drivers impact the optimal quality level and observe some intriguing phenomena. We find that the product quality does not necessarily increase as either the high segment is willing to pay more or there are relatively more high-segment consumers. Additionally, the impacts of returns policies are ambiguous from the perspectives of either social welfare or consumer surplus. When instead the low-segment consumers are associated with a higher hassle cost or the residual value is sufficiently high, the manufacturer has no incentive to offer returns, irrespective of the consumers' population and the quality risk.

We organize the rest of this article as follows. The relevant literature is reviewed in section 2. In section 3, we lay out the model. Section 4 derives the optimal pricing and returns policies when the product quality is exogenously given, and section 5 discusses the scenario in which quality design is endogenous. In section 6, we investigate some variants of our model characteristics. Section 7 provides some concluding remarks and research extensions. All the proofs are relegated to the Appendix S1.

2. Literature Review

Our article is related to a long-standing literature on product returns. In Davis et al. (1995), consumers purchase products that either match the consumers' tastes or not (as we do in our model). In the direct marketing context, Hess et al. (1996) investigate how the manufacturer can use non-refundable shipping/handling/restocking fees to eliminate consumers' "inappropriate returns," that is, avoid those who purchase the products with no intention of keeping them; see also Chu et al. (1998). Davis et al. (1998) study the role of lenient and stringent returns policies and the impact on consumers' hassle (returns) costs. In contrast with the aforementioned papers that focus exclusively on full refunds (or no refund), we allow for general returns policies and demonstrate the profitability of excessively generous returns policy in various scenarios. Furthermore, we investigate the interplay between product quality, pricing strategy, and the returns policy, which has no counterpart in the existing literature to the best of our knowledge. It is also worth mentioning that in our article, the manufacturer finds it optimal to prevent inappropriate returns from consumers with no intention of keeping their purchases, even if the consumers exhibit heterogeneous (uncertain) valuations and hassle costs *ex ante*. Padmanabhan and Png (1997) investigate a distribution channel in which the manufacturer sells through two competing retailers. They argue that with demand uncertainty, the returns policy may mitigate the retailers' competition, thereby benefiting the manufacturer; see also Wang (2004) and Padmanabhan and Png (2004) for some follow-up discussions. Unlike these papers, we provide a micro-foundation of the consumer behavior by explicitly modeling their utility functions and decision making.

Operations researchers have also investigated how returns/refund policies help mitigate consumers'

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concerns when they exhibit valuation uncertainty (e.g., Akan et al. 2009, Liu and Xiao 2008, Su 2009, Swinney 2011). Akan et al. (2009) show that when consumers observe their true valuations at different time epochs, the manufacturer can design appropriate menus of expiring returns for which the prices and refunds change over time. Liu and Xiao (2008) incorporate the capacity constraint into the model investigated by Akan et al. (2009) and show that the menu of returns policies is no longer optimal and proceed to evaluate three different selling policies commonly used in practice. Su (2009) studies the impact of consumer returns on supply chain performance when the consumers exhibit valuation uncertainty and the manufacturer is confronted with demand uncertainty. Swinney (2011) examines how the firms should design the returns policy to induce early/premature purchases when consumers are only partially aware of their true valuations. He further shows that the quick response system may be detrimental to the firm's profit although it does mitigate supply-demand mismatch. See also Dana (1998) and Xie and Shugan (2001) for more discussions on the impact of consumer valuation uncertainty. In the aforementioned papers, all consumers who return the products to the store uniformly incur the same hassle cost (may be normalized to zero for simplicity), whereas our article explores the economic consequence of heterogeneous hassle costs.

As the product returns serve as an instrument to protect consumers against product and/or valuation uncertainties, our article is also related to the large literature on product warranties. The conventional view for offering generous returns is to signal the product quality when the consumers face the adverse selection—the lemons market problem. In the classical paper, Moorthy and Srinivasan (1995) demonstrate that although offering generous returns is costly to the sellers, it is relatively more costly for the sellers whose product quality is low. This provides an effective tool for high-quality sellers to distinguish themselves from others, see also Emons (1989), Grossman (1981), and Shieh (1996) for similar arguments. There are also papers that demonstrate the insurance effect (Gu and Tayi 2010, Heal 1977, Matthews and Moore 1987, Padmanabhan and Rao 1993, Welling 1989), and the central premise is that a more generous policy boosts consumer demand and increases the sellers' gross revenue. These two research streams seem to suggest that an online seller with a higher reputation has a stronger incentive to provide very generous returns policy, presumably covering the shipping costs for returns and eliminating the restocking fee. In contrast, our results reveal the possible non-monotonicity between the product quality and the refund amount. It is worth mentioning that in Spence (1977), the presence of product returns allows the manufacturer (seller) to credibly convey information regarding the product quality to the consumers, thereby providing the desired incentive for the manufacturer to invest on product quality *ex ante*. Our results complement this classical result by showing that in the absence of the adverse selection problem, the flexibility of offering consumer returns may *deter* the manufacturer from quality improvement.

3. Model

We consider a stylized model in which a monopolistic manufacturer intends to sell his products to consumers with heterogeneous preferences. To model the potential product defect/misfit/inappropriateness, we use α to denote the probability that the product "works normally" (i.e., meets the consumers' expectations) and $1 - \alpha$ thus represents the probability that an inappropriate product appears. As aforementioned, inappropriate products may arise from product misfit, product defect, or product unconformity. This notion of misfit probability coincides with Davis et al. (1995, 1998) and Hess et al. (1996), where they also describe the product mismatch phenomenon by a binary probability distribution. Initially, we assume that the quality is exogenously given, presumably because of the product development stage, the contemporary technological frontier, the inability to improve upon current product description. Later, in section 5, we investigate the scenario in which the quality of the product is the manufacturer's discretion.

3.1. Consumers

Each consumer intends to purchase (at most) one unit of the product, and exhibits heterogeneous valuations upon using a product that works normally. To provide the simplest possible ground for this heterogeneity, we assume that there are two consumer segments, the low and high segments. Consumers within each segment are homogeneous. A low-segment consumer obtains valuation v_1 from using the "perfectly functioning" (appropriate) product, whereas a high-segment consumer obtains valuation v_2 ($0 < v_1 < v_2$). This vertically differentiated two-segment model is also adopted in numerous papers, for example, Gerstner and Holthausen (1986), Lazear (1986), Pashigian (1988), Raju et al. (1990), and Varian (1980). In our basic framework, an inappropriate product generates no utility for the consumers, regardless of which segment she belongs to. This assumption is made for simplicity (see, e.g., Chu et al. 1998, Davis et al. 1995, 1998, Hess et al. 1996, Moorthy and Srinivasan 1995), as this extreme case clearly demonstrates the difference of consumer preferences between the appropriate

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and inappropriate products. In section 6, we extend our analysis to investigate the case when the consumers obtain residual valuations from inappropriate products. The proportions of the low and high segments are, respectively, $1 - q$ and q, where $q > 0$.

3.2. Returns

Given the possibility of inappropriate products, consumers may be hesitant to purchase the product in the first place. To overcome such a difficulty, a natural solution is to offer returns to the consumers. In addition to the valuation heterogeneity, the consumers are also heterogeneous in their costs of returning the products. As we are primarily interested in the returns policies in clicks-and-mortar e-business, we interpret these hassle costs as the heterogeneous costs of keeping the products and their time values. In our basic framework, we assume that high-segment consumers are endowed with a higher hassle cost than the low-segment consumers, and we use T and t to denote these costs for the high and low segments, respectively, where $0 < t < T$. This assumption is intended to capture the fact that high-segment consumers are willing to pay more for the product because they are wealthier; thus, they have a higher opportunity cost of spending their time and effort elsewhere (see, e.g., Anderson and Song 2004, Gerstner and Hess 1991, 1995). If instead the hassle costs correspond to the different transportation costs to the physical stores in the offline (bricks-and-mortar) world or the money-tied-up issue, it is possible that low-segment consumers incur a higher monetary hassle cost than the high-segment consumers do. This alternative scenario is discussed in section 6. Upon receiving the returned merchandise, the manufacturer receives some salvage value, denoted as s. We assume that $0 < s < t$, so that returns are inefficient in the sense that it generates a net negative social surplus. Moreover, we normalize the reservation utility for the consumers to zero, regardless of whether she belongs to the high or low segment.

3.3. Manufacturer's Goal

The manufacturer's goal is to maximize his expected payoff taking into account the potential inappropriate products and consumer returns. To this end, he has to determine the selling price, denoted by p, and the refund upon receiving a consumer return, denoted by f. Specifically, if $f = 0$, then effectively no returns are allowed. In contrast, if $f = p$, consumers are entitled for a full refund upon returning an inappropriate product. From our general setup, a partial refund is also allowed ($0 < f < p$) (except for product defects), and the manufacturer can over-compensate the consumers ($f > p$) if he finds it profitable to do so (which corresponds to the "satisfaction guaranteed policy" in Fruchter and Gerstner 1999, Moorthy and Srinivasan 1995).

3.4. Timing

The sequence of events proceeds as follows: (a) the manufacturer determines the selling price p and the refund f; (b) each consumer decides whether to purchase the product; (c) if a consumer purchases the product, she decides whether to return the product after figuring out whether an inappropriate product appears. As in reality, it is typically difficult to verify the inappropriate products right at the spot when the products get returned, our model allows a consumer to return a perfectly functioning product if the returns policy is too generous. This may arise, for example, when product defects require detailed inspections, or when the product misfit/unconformity comes entirely from the consumers' subjective perception. In Figure 1, we demonstrate the possible flow of the game sequence.

3.5. Manufacturer's Strategies

As the high-segment consumer obtains a higher valuation and incurs a higher hassle cost, *ex ante* it is unclear whether the manufacturer prefers to accommodate the high segment or the low segment. Thus, in the sequel, we will examine three possible strategies, and for ease of exposition we label them as follows: (a) strategy H: inducing the high segment to purchase the product and excluding the low segment; (b) strategy L: inducing the low segment to purchase the product and excluding the high segment; and (c) strategy B: inducing both segments to purchase (B stands for "both"). In strategy B, we further distinguish between selling to both segments without returns (BN) and selling to both segments with returns (BR). Note that in our framework, we segment consumers by their endowed valuations (v_1 and v_2) and interpret q as the *ex ante* proportion of consumers who attach valuation v_1 to an appropriate product. This parameter q corresponds to the consumers' intrinsic attributes. However, as we use the mismatch probability α to account for the likelihood that a

Figure 1 Flow of the Game Sequence



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consumer may find the product less desirable, this probability is attached directly to the product characteristics. Therefore, these primitives should not depend on the manufacturer's pricing strategy and return policy. However, these strategic decisions do affect the *ex post* segmentation of consumers in terms of their purchasing and returning behaviors, as indicated in the above strategies H, L, and B.

As the game involves multiple rounds of strategic interactions, we use subgame perfect Nash equilibrium as our solution concept. In the next section, we characterize the optimal pricing strategy and the returns policy for an exogenously given product quality. Following this, we proceed to examine the situation in which the manufacturer can design the product quality in section 5.

4. Optimal Returns Policy and Pricing Strategy

In this section, we characterize the optimal returns policy and pricing strategy for an exogenously given quality. Following this, we derive some managerial implications from our suggested policies.

4.1. Equilibrium Strategies

We use backward induction to characterize the equilibrium and provide some detailed derivations of consumer decision making, which illustrates how the pricing strategy and return policy affects consumers' purchasing behaviors and later on will become the core concern for the manufacturer.

4.1.1. Consumers' Return Strategies. We shall first assume that a consumer has purchased the product and has figured out whether the inappropriate product occurs. At this moment, her only decision is whether to return the product to the manufacturer. If the product works well, a low-segment consumer chooses to return it if and only if $f - t > v_1$, whereas a high-segment consumer returns whenever $f - T > v_2$. In contrast, when the inappropriate product appears, a low-segment (high-segment) consumer returns the product if and only if $f > t$ ($f > T$).

4.1.2. Consumers' Purchasing Decisions. Now let us return to the second stage and evaluate whether a consumer is willing to purchase the product. To illustrate, consider how a low-segment consumer makes her purchasing decision. When $f > v_1 + t$, a low-segment consumer returns the product afterwards regardless of whether it works; thus, a low-segment consumer purchases the product if and only if $f - t - p \geq 0$, where 0 corresponds to her reservation utility for not purchasing the product. When $t < f \leq v_1 + t$, after purchasing the product, a low-segment consumer only returns the inappropriate one. Thus, a low-segment consumer obtains an expected payoff $\alpha(v_1 - p) + (1 - \alpha)(f - t - p)$, where the first term corresponds to the case when the product functions well, and the second term indicates the scenario with an inappropriate product (and the consumer returns the product for refund f). In this case, a low-segment consumer purchases the product if and only if $\alpha(v_1 - p) + (1 - \alpha)(f - t - p) \geq 0$. Finally, when $f \leq t$, a low-segment consumer never returns the product. Thus, her expected payoff upon purchasing the product is $\alpha v_1 - p$, and she purchases whenever this expected payoff is positive. Similarly, we can also characterize the conditions under which a high-segment consumer purchases the product as follows.

4.1.3. Manufacturer's Problem. Regarding the manufacturer's problem in the first stage, it is conceivable that the formulation crucially depends on which strategy the manufacturer intends to implement, or equivalently, which region the refund f falls into. As an example, if the refund is so low that no returns takes place and the manufacturer sells to both segments (strategy B), his expected payoff is the difference between the selling price and the quality cost:

$$\pi_M = \max_{p \geq 0} \{p - k\alpha^2 \mid \text{s.t. } p \leq \alpha v_1\},$$

where the constraint ensures that the low-segment consumers are willing to purchase.

In contrast, if the manufacturer intends to sell to both segments but only induce the low segment to return the inappropriate products, his expected payoff is as follows:

$$\pi_M = \max_{p \geq 0, f \geq 0} \{p - (1-q)(1-\alpha)(f-s) - k\alpha^2 \mid \text{s.t. } \alpha(v_1 - p) + (1-\alpha)(f - t - p) \geq 0 \text{ and } p \leq \alpha v_2\},$$

where in the objective $(1 - q)(1 - \alpha)$ is the probability that a low-segment consumer appears and returns the product, in which case the manufacturer refunds her f and salvages the product at value s. The two constraints guarantee that both segments are willing to purchase, where the right-hand sides correspond to their expected payoffs for the low and high segments, respectively. Naturally, there are many other cases in which the manufacturer plans to implement different pricing strategies. Although we relegate all the detailed formulations and analysis to Appendix S1, the results are summarized in the following proposition.

PROPOSITION 1. *Suppose that $T > t$. At optimality, when returns are offered, the manufacturer sells to both*

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segments and induces the low-segment consumers to return the inappropriate products.

Proposition 1 identifies the intrinsic role of returns in our context. Specifically, when the consumers exhibit heterogeneous hassle costs, the manufacturer uses returns to implicitly distinguish the consumer segments through their self-selection behaviors. When the returns are accepted, the manufacturer can wisely set an appropriate refund so that the high-segment consumers are unwilling to honor the policy even if the product turns out to be inappropriate; on the contrary, because of their low hassle cost, the low-segment consumers are encouraged to return the inappropriate products. This facilitates the tacit discrimination among consumers by elaborating on their heterogeneous hassle costs. Moreover, the manufacturer never finds it optimal to induce the low-segment consumers to return the product that functions well, even if the manufacturer is entitled to charge nonrefundable restocking fees. This is because the returns are inefficient from the society's perspective (as the consumers' hassle cost is unambiguously higher than the manufacturer's salvage value). Consequently, as long as the returns policy serves well the price discrimination purpose, the manufacturer intends to minimize the deadweight loss by inducing the low-segment consumers to only return the inappropriate product.

Furthermore, even if the manufacturer is unable to inspect the inappropriate product at the moment of returns, he can design the returns policy so that products that function well never get back to the store. Although we do not model the *ex post* inspection and repackaging behind the scenes, our suggested returns policy may allow the manufacturer to bypass such an operational burden because all the returned products are necessarily inappropriate. This may be particularly useful with product defects/breakdowns. Naturally, this result holds under our specific model characteristics, and it highly depends on the assumption that the manufacturer has perfect knowledge regarding the consumer segmentation. Our analysis may help the manufacturer to determine how much effort he should put in understanding the fine details of consumer heterogeneity. We also observe that as the returns are inefficient in our context, it may be optimal for the manufacturer to abandon the returns in certain scenarios, for the benefit of price discrimination may be outweighed by the inevitable deadweight loss.

Proposition 1 implies that there are three possible scenarios that emerge as the optimal strategy: (a) selling only to the high segment (H), (b) selling to both segments without returns (BN), and (c) selling to both segments with returns (BR). An immediate question is "When will each scenario occur?" To answer this question, we compare the corresponding expected profits of the manufacturer in different scenarios and conduct some comparative statics on the primitives; the results are summarized in Table 2. Although this table provides a comprehensive study of all relevant factors, let us focus primarily on the effects of the product quality (α) and the valuation of the low segment (v_1) and use a numerical example to graphically illustrate them in Figure 2.[1] From Figure 2 and our analytical results, we observe that

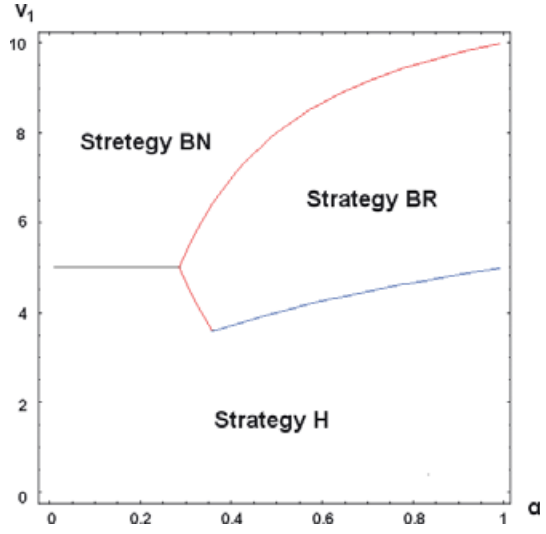
COROLLARY 1. *Suppose that $T > t$. For any given probability α, there exist a pair of thresholds $v_{1a}(\alpha)$ and $v_{1b}(\alpha)$ such that strategy BR is optimal when $v_{1a}(\alpha) \leq v_1 \leq v_{1b}(\alpha)$. Likewise, if we fix a valuation v_1, there exist a pair of thresholds $\alpha_a(v_1)$ and $\alpha_b(v_1)$ such that strategy BR is optimal when $\alpha_a(v_1) \leq \alpha \leq \alpha_b(v_1)$.*

Corollary 1 shows that consumer returns are offered only when the product quality is moderate and the valuation of the low segment is moderate. This could potentially provide an answer to why we observe various return policies among Amazon, Famous Footwear, OnlineShoes, Payless, RoadRunner Sports,

Table 2 The Manufacturer's Optimal Strategies in Different Scenarios

	Strategy BR	Strategy H	Strategy BN
α	Moderate	Large	Small
v_1	Moderate	Small	Large
v_2	Moderate	Large	Small
q	Moderate	Large	Small
$T - t$	Large	Small	Small
s	Large	Small	Small
k	Small	Large	Small

Figure 2 Optimal Strategies vs. Product Reliability and Low-Segment Valuation



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Shoebacca, ShoeBuy, ShoeMall, Shoes.com, although they sell similar products—shoes. These results may be explained as follows. Note that the gross valuations for obtaining the product (excluding the subsequent returns) are αv_2 and αv_1 for the high and low segments, respectively. When the product quality is very high (α is close to 1), the high segment is much more profitable than the low segment from the manufacturer's perspective. Consequently, the manufacturer intends to set a high selling price to target the high segment. In this way, the manufacturer can effectively exclude the low segment, whereas by allowing returns, the manufacturer has to deal with the cannibalization problems. On the other extreme, when the product quality is relatively low (α is close to 0), if the returns were offered, the manufacturer would have to largely compensate the low segment for returning to the store. As this is inefficient and the manufacturer's salvage value is relatively low, this over-compensation becomes unprofitable for the manufacturer. In such a scenario, either targeting the high segment (when v_1 is small) or selling to both segments without returns (when v_1 is large) is more beneficial to the manufacturer than offering returns. In the intermediate range, returns can then serve as an effective and profitable screening tool that differentiates the high and low segments. Note that in some degenerate cases, these thresholds may coincide with each other and $\alpha_b(v_1) = 1$ may be possible, as we demonstrate in Figure 2.

Now let us articulate the impact of the low-segment consumers' valuation (v_1) on the manufacturer's optimal strategy. When the valuation of low-segment consumers is relatively low, the manufacturer finds it unprofitable to serve the low segment; consequently, he sets a high selling price and targets only the high segment. However, when the low-segment consumers have a relatively high valuation, they become increasingly important from the manufacturer's perspective. If returns are offered, the manufacturer has to compensate the low-segment consumers for bringing back the inappropriate products; consequently, these inefficient returns reduce the profitability for serving the low segment. Thus, it is optimal for the manufacturer to simply set a low selling price to sell to both segments and abandon the (inefficient) returns, that is, strategy BN now dominates strategy BR.

4.2. Implications

Next, we provide some implications based on our analytical results.

4.2.1. Social Welfare. We now articulate the impacts of returns policies on social welfare and consumer surplus, which are summarized in Table 3. From this table, we find that the impacts are ambiguous from the perspectives of either social welfare or consumer surplus, and they crucially depend on whether originally the manufacturer intends to serve the entire market. Specifically, suppose that the manufacturer originally serves only the high segment (strategy H) but now offers returns to the consumers (strategy BR). In this case, the high-segment consumers now enjoy information rent as the manufacturer is unable to distinguish between the low and high segments; on the other hand, the low-segment consumers receive a null payoff as before. Furthermore, switching to strategy BR must be profitable for the manufacturer. Collectively, both social welfare and consumer surplus increase as the manufacturer offers returns to consumers. Now suppose that the manufacturer originally serves both segments (strategy BN) and starts to offer returns. In this case, the high-segment consumers' surplus decreases because returns serve as an effective price discrimination instrument, whereas the manufacturer gains by extracting more revenue from the consumers. Overall, social welfare decreases as returns are socially wasteful.

4.2.2. Product Quality. We can also examine how the product quality affects the manufacturer's pricing strategy and the refund amount (provided that returns are offered). In Figures 3 and 4, we plot the selling price, the refund, and the ratio f/p vs. the product quality (α) for different hassle costs. In Figure 3, the corresponding parameters are as follows $v_2 = 3$, $v_1 = 2.5$, $T = 2$, and $t = 1$. In Figure 4, we use $v_2 = 3$, $v_1 = 1.5$, $T = 3$, $t = 0.2$. Note that the ratio depicts how generous the returns policy is. For example, if $f/p < 1$, the manufacturer offers only a partial return and $(1 - f/p) \times 100\%$ can be regarded as the non-refundable charge or the restocking fee. When $f/p = 1$, consumers are entitled to a full refund. When $f/p > 1$, the manufacturer offers a very generous returns policy as he intends to compensate the consumers for returning

Table 3 Social Welfare and Consumer Surplus

Strategy	BR	BN	H
Optimal price	$\alpha v_1 + (1 - \alpha)(f^* - t)$	αv_1	αv_2
High segment's surplus	$\alpha(v_2 - v_1) - (1 - \alpha)(f^* - t)$	$\alpha(v_2 - v_1)$	0
Low segment's surplus	0	0	0
Manufacturer's expected profit	$\alpha v_1 - k\alpha^2 + q(1 - \alpha)(f^* - t) - (1 - q)(1 - \alpha)(t - s)$	$\alpha v_1 - k\alpha^2$	$q(\alpha v_2 - k\alpha^2)$
Social welfare	$q\alpha(v_2 - v_1) + \alpha v_1 - k\alpha^2 - (1 - q)(1 - \alpha)(t - s)$	$q\alpha(v_2 - v_1) + \alpha v_1 - k\alpha^2$	$q(\alpha v_2 - k\alpha^2)$

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Figure 3 Pricing Strategy and Returns Policy vs. Product Quality (When $T \leq v_1 + t$)

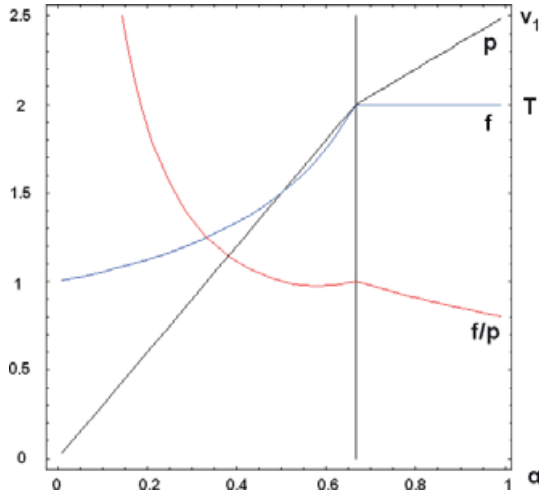
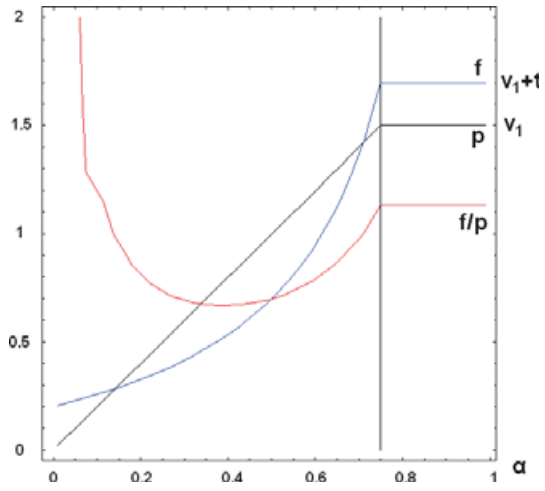


Figure 4 Pricing Strategy and Returns Policy vs. Product Quality (When $T > v_1 + t$)



the (inappropriate) product to the store. The results are summarized in the next corollary.

COROLLARY 2. *Suppose that $T > t$ and the manufacturer sells to both segments and allows returns (strategy BR). We find that*

1. *When the product quality is relatively low, the manufacturer may over-compensate the consumers for returning the product ($f/p > 1$).*
2. *When the product quality is relatively high, further improvement in product quality may not induce a higher selling price (and the corresponding refund).*
3. *The ratio f/p may be non-monotonic in product quality.*

Corollary 2 lists some interesting properties of the optimal pricing strategy and returns policy. To understand the first result, observe that when the product quality is very low, the selling price should be set low to ensure participation from the low segment. However, for the price discrimination purpose, the manufacturer must induce the low-segment consumers to return the inappropriate product; consequently, the refund should compensate their hassle cost and thus is lower bounded by t. This explains why when the product quality is relatively low, the manufacturer intends to offer a seemingly overly generous returns policy. Notably, this strategy corresponds to the "satisfaction guaranteed" concept proposed by Fruchter and Gerstner (1999) and speculated in Moorthy and Srinivasan (1995, section 6), and is in line with the current practice by Endless.com, PiperLime.com, Shoebacca.com, ShoeBuy.com, Shoes.com, and Zappos.com. Second, when the product quality is relatively high, we observe that the selling price and/or the refund may flatten out as product quality improves. This is because in this regime, the returns policy must induce voluntary separation between two segments. Therefore, the refund is set either to match the high-segment consumers' hassle cost ($f = T$) or just enough to deter the low-segment consumers from returning the products that function well ($f = v_1 + t$). Given this, further improvement on product quality does not boost a higher expected profit. Our results thus identify an intrinsic rationale for the manufacturer's aversion to technological improvement in the context of consumer returns. Incidentally, the possibility of partial returns documented in our setting may explain why in the presence of quality risk, Amazon.com, OnlineShoes.com, and ShoeMall.com do not offer free returns.

4.2.3. Restocking Fee. Next, we observe that f/p, the proxy of restocking fee, may be non-monotonic in product quality. Given that the refund is lower bounded, it is not surprising that when the product quality is relatively low the ratio is high. On the other extreme, as the refund is upper bounded by either T or $v_1 + t$, a decreasing pattern can also be expected. A rather subtle point is that in the intermediate region, the manufacturer is increasingly generous in reducing the restocking fee, as the ratio is monotonically increasing in product quality. This implies that the refund amount grows faster than the selling price. This may be explained as follows. As the product quality becomes higher, the manufacturer naturally intends to set a high selling price as the intrinsic quality becomes higher. However, to induce the low segment to purchase the product, she has to compensate their hassle cost by offering an appropriate refund. When the product quality is higher, the chance of

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returning the product becomes lower. Thus, the manufacturer must largely increase the refund amount, as the low-segment consumers honor the returns less often and their participation crucially depends on this refund. Collectively, we observe that the refund amount may increase at a faster rate than the selling price as the product quality becomes higher.

4.2.4. Full Return Restriction. In the above discussions, we impose no restriction on the restocking fee. Nevertheless, in the context of product defects, the manufacturer may be regulated to provide (at least) a full return. This is possible when the manufacturer provides the communication channel for the consumers to provide the evidence and the evidence can potentially be validated by the court. In such a scenario, we can conveniently add an additional constraint $f \geq p$. This allows us to identify the region in which a full return is offered, as we summarize in the next corollary.

COROLLARY 3. *In the context of product defects, the manufacturer offers full returns when*

$$\frac{v_1 + t - \sqrt{(v_1+t)^2 - 4tv_2}}{2v_2} \leq \alpha \leq \min\left\{\frac{\min\{v_1+t, T\} - t}{v_2 - v_1 + \min\{v_1+t, T\} - t}, \frac{v_1 + t + \sqrt{(v_1+t)^2 - 4tv_2}}{2v_2}\right\}$$

or

$$\alpha > \max\left\{\frac{T-t}{v_2 - v_1 + T - t}, \frac{t}{v_1 + t - T}\right\}, \quad \textit{and } T \leq v_1 + t.$$

Corollary 3 shows that in the context of product defects, full returns are offered when the product quality is not too low. With these explicit conditions, interested readers can easily derive the comparative statics and managerial implications of the return policy for product defects.

5. Optimal Choice of Quality Risk

In this section, we investigate the situation in which the quality risk is the manufacturer's discretion. As aforementioned in the introduction, this quality choice is applicable only to the cases of product defect and product unconformity. In the case of product misfit, the manufacturer is unable to unambiguously reduce the probability of mismatch because of the intrinsic heterogeneity of consumer tastes. In such a scenario, the manufacturer incurs a cost of $C(\alpha) = k\alpha^2$ for producing one unit of a product with quality α, where $k > 0$ and the quadratic form is made for simplicity. To ensure that at optimality the probability is reasonably bounded ($0 < \alpha < 1$), we assume that $k > v_2/2$. This essentially rules out the possibility of corner solutions for which the manufacturer intends to produce a product that is necessarily appropriate, leaving the returns policy meaningless. Note that in principle, we shall distinguish between the case with variable (marginal) costs and fixed costs in the product quality improvement. However, in our context, this distinction does not affect the returns policy, since in section 4, we have shown that returns are offered only when both segments purchase the products. Given this observation, the marginal costs are incurred for all the units sold to the consumers, and equivalently these can be taken out as a lump-sum amount that affects the manufacturer's returns decision in exactly the same way as a fixed cost.

While in the Appendix S1 we work out the detailed analysis for all possible scenarios, to crystallize the manufacturer's incentive for quality design and its interplay with the returns policy, we shall, in the main text, focus exclusively on the scenario in which the manufacturer attempts to serve both segments and returns are allowed (i.e., the BR strategy). The discussions for the other two scenarios (selling to both segments without returns or selling to only the high segment) are rather straightforward and thus are omitted.

To facilitate our discussions, we define three different regions depending on the hassle cost differential between the high and low segments (i.e., $T - t$), in descending order. For ease of exposition, the discussions in the sequel focus exclusively on the case in which $T - t < v_1$; this essentially implies that the benefit of serving the low segment is relatively high as the hassle cost differential is low. All the results are qualitatively similar for the alternative case $T - t \geq v_1$. In Regions I and II, the manufacturer intends to extract the entire surplus from both segments. Thus, the selling price is set at the expected gross utility of the high segment (αv_2), and the corresponding refund is designed to induce the low segment to purchase the product and prevent the high segment to return the inappropriate product. The only difference between these two regions is the effectiveness of price discrimination: it is highly effective

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in Region I as the two segments incur very different hassle costs. When the two segments experience similar return costs in Region III, the manufacturer is unable to extract the full surplus from both segments; rather, his primary concern is how to segment the market (by preventing the high segment from returning the product). In Table 4, we demonstrate the conditions in which these regions are more likely to arise.

Let us now investigate the driving forces of the optimal quality from the manufacturer's perspective. While Table 5 summarizes the comparative statics of the optimal quality level, we shall only selectively discuss some intriguing findings. First, we observe that the product quality does not necessarily increase as the high segment is willing to pay more (when v_2 increases). This may be explained as follows. When the hassle cost differential is large (Region I), the manufacturer is able to extract full surplus from the high segment by setting a high selling price and offering the returns to compensate the low segment. Thus, as the high segment exhibits a higher valuation, the manufacturer intends to improve the product quality due to the multiplicative effect. However, when the hassle cost differential is moderate, the manufacturer attempts to decrease the product quality to facilitate price discrimination in response to an increase of high segment valuation (v_2). When the hassle cost differential becomes sufficiently small, the manufacturer is less concerned about extracting surplus from the high segment; rather, his primary goal is to obtain the maximum benefit from the low segment. Consequently, the pricing strategy and the corresponding refund policy are designed entirely for inducing the low segment to participate. Thus, in this region, the quality design problem is independent of the high-segment valuation. Similar arguments can be applied to explain the non-monotonicity of product quality with respect to the return costs for high and low segments, and we omit the details here to avoid redundancy.

We also observe that, even when price discrimination is very effective (Region I), the quality is not necessarily increasing in the population of high-segment consumers (q). To understand this result, it is helpful to elaborate on how the manufacturer collects the revenue from the high and low segments. In Appendix S1, we show that the gross revenues from the high and low segments are, respectively, αv_2 and $\alpha v_1 - (1-\alpha)(t-s)$, where $(1-\alpha)(t-s)$ corresponds to the manufacturer's expected loss via returns. The improvement of product quality gives rise to two benefits for the manufacturer from serving the low segment. On one hand, he is able to extract more gross revenue from the low segment (via αv_1); on the other hand, he can also save the loss as the returns become less likely. This effect is particularly pronounced as $t-s$, the deadweight loss upon returns, is sufficiently large. Consequently, these two effects collectively affect how the change of consumer population determines the manufacturer's incentive to improve product quality. When the aggregate gain from the low segment dominates that from the high segment, it is possible that a higher proportion of high-segment consumers leads to a reduction of product quality.

Table 4 The Likelihoods of Three Regions ($T - t < v_1$)

	Region I	Region II	Region III
v_2	Small	Moderate	Large
v_1	Large	Moderate	Small
T	Large	Moderate	Small
t	Small	Moderate	Large
q	Ambiguous	Ambiguous	Small
s	Large	Moderate	Small
k	Large	Moderate	Small

Table 5 Comparative Statics for the Optimal Quality ($T - t < v_1$)

	Region I	Region II	Region III
$\partial\alpha^*/\partial v_2$	+	−	0
$\partial\alpha^*/\partial v_1$	+	+	+
$\partial\alpha^*/\partial T$	0	+	−
$\partial\alpha^*/\partial t$	+	−	+
$\partial\alpha^*/\partial q$	ambiguous	0	−
$\partial\alpha^*/\partial s$	−	0	−
$\partial\alpha^*/\partial k$	−	0	−

6. Alternative Scenarios

In this section, we consider some variants of our model characteristics that are more in line with the offline (brick-and-mortar) channels. Our goal is to investigate how these factors significantly impact the seller's incentive to offer returns to the consumers.

6.1. When the High-Segment Consumers Incur a Lower Hassle Cost

In our basic framework, we assume that high-segment consumers incur a higher hassle cost for returning the products, as in the online world the hassle costs primarily arise from the consumers' time values. In contrast, in the offline world, the majority of hassle costs is attributed to the actual transportation costs incurred by the consumers for visiting the physical stores. In such a scenario, it is possible that low-segment consumers may incur a higher hassle cost as low-income families may be forced to spend more in transportation and living. This somewhat counterintuitive phenomenon is actually quite common nowadays. As indicated in a recent empirical study by Talukdar (2008), poor people (low-segment consumers in our context) typically "face a double squeeze in their consumption experience" because of the

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common inflexibility and inconvenience they incur in the daily lives. Specifically, poor people have limited access to services, public transportation, and typically live in rural areas due to the limited ability to pay. Also, a study by the Annie E. Casey Foundation shows that "[i]n rural areas, public transportation is rarely an option" for those low-income people; thus, 98% of rural working families have to rely on their personal cars but owning a car is also too expensive for many of them (see "The high cost of being poor" at http://www.aecf.org/). Consequently, their costs of collecting and using coupons (through, e.g., collecting them on the Internet, sending them in the post office, going to the original retail store) are also higher than those of the richer people. Notably, this phenomenon may occur even in the absence of transportation issues. In particular, the money tied up through the purchase of the product can be a significant concern for a consumer, as receiving the refund can take up to a couple of weeks. This may not be an issue for high-segment consumers who have additional funds, but it could significantly affect the low-segment consumers.

To accommodate this possibility, we in the following assume that the low-segment consumers are associated with a higher hassle cost. With some abuse of notation, we use T and t to denote the hassle costs of the low-segment and high-segment consumers, respectively, where we recall that $0 < t < T$. We find that in this alternative scenario, the manufacturer has no incentive to offer returns to the consumers.

PROPOSITION 2. *When the low-segment consumers have a higher hassle cost, the manufacturer prefers not to offer returns to the consumers, irrespective of the consumers' population and the quality risk.*

To understand this result, recollect that in the presence of consumer heterogeneity, the primary goal of returns policy is to facilitate discrimination among the consumers through their self-selection behaviors. To this end, the manufacturer should set a high selling price to extract the revenue from the high-segment consumers, and compensate the low-segment consumers via generous refunds. This instrument fails to serve its purpose when the low-segment consumers incur a higher hassle cost, because in this case, the high-segment consumers will necessarily be more tempted to return the inappropriate products than the low-segment consumers. Consequently, the manufacturer loses the chance to price discriminate among the consumers and is better off not accepting any returns.

6.2. When Inappropriate Products Yield High Residual Values

In our basic framework, we assume that inappropriate products do not yield any value for the consumers to facilitate the analysis. This assumption is directly applicable to the product defect case, but is rather restrictive for the cases with product misfit or product unconformity, as the consumers may still be able to use the products despite the mismatch. To incorporate this possibility, we now amend our framework and assume that consumers in segment i obtain utility βv_i from using an inappropriate product, where $0 \leq \beta < 1$ captures the residual value of the products that do not function well. Our basic framework thus corresponds to the special case where $\beta = 0$.

As an interpretation of this extended model, we can suppose that the quality level of the appropriate product as normalized to 1, and β denotes the quality level of the inappropriate product. Accordingly, v_1 and v_2 correspond to the consumers' willingness to pay for quality as in the product line design literature (Moorthy 1984, Mussa and Rosen 1978). Alternatively, one can also interpret $1 - \beta$ as the *perceived* quality difference between appropriate and inappropriate products from the consumers' viewpoints, and it measures the quality uncertainty faced by the unsure consumers when they purchase the products. Since consumers who intend to engage in Internet transactions are unable to actually see and inspect the products before making their purchases, this quality uncertainty shall be more significant in the clicks-and-mortar transactions than the bricks-and-mortar ones. Viewed in this way, a lower value of β brings us closer to the online world, whereas a higher β is more in line with the traditional offline transactions.

Since the consumers' preferences, and therefore the manufacturer's expected payoff, are continuous in the parameter β, it is conceivable that all the results in sections 4 and 5 hold qualitatively when β is close to 0. Thus, in the following, we focus instead on the other end of the spectrum. Our results show that if the quality uncertainty is sufficiently low (β is large), the manufacturer finds it unprofitable to offer the returns.

PROPOSITION 3. *When $\beta > \max\{v_1/v_2, q\}$, the manufacturer prefers not to offer returns to the consumers, irrespective of the quality risk and the hassle costs.*

Note that an increase of β leads to two effects. On one hand, as the inappropriate products yield higher residual values for the consumers, it is socially more desirable to keep the products in the consumers' hands to avoid the wasteful hassle costs. This implies that the rationale for offering returns to improve social efficiency goes down. On the other hand, a higher β also effectively reduces the preference difference between the two consumer segments, as the valuation differential decreases when the quality uncertainty drops. Specifically, the perceived valuation differentials between the appropriate and

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inappropriate products from the consumers' viewpoints are $(1 - \beta)v_1$ and $(1 - \beta)v_2$, respectively. Thus, the incentive for the manufacturer to differentiate between these two segments declines as β becomes higher, thereby reducing the profitability of offering returns for the price discrimination purpose. As both effects work against returns, the manufacturer ultimately abandons the returns policy when β is sufficiently large. We further find that this intention is stronger as the manufacturer is less concerned about the low-segment consumers, either when their valuation becomes lower (v_1/v_2 becomes small) or there are only a few of them (q is small). This is because when the low-segment consumers are relatively less financially influential, it does not pay back the manufacturer to compensate them via the generous refunds.

7. Conclusions and Discussions

In this section, we summarize the similarities and differences of the three notions of quality risks, discuss our analytical results, and provide some possible extensions.

7.1. Categories of Quality Risks

In this article, we define quality risk as the possibility of product misfit, defect, or unconformity with the consumers' perception. These notions of quality risks differ in return policy restriction, residual values, and whether it is possible to unambiguously reduce the probability of mismatch; see Table 6 for a detailed categorization. It is worth mentioning that in practice most of the manufacturers and retailers do not inspect the returned products right on the spot. For various leading retailers, they even employ a "no-questions-asked" policy. On one hand, it does not require the consumers to provide any evidence for the legitimate reason of returns. On the other hand, however, in the absence of real-time, scientific inspections, the reasons for returns inevitably incorporate the *subjective* assessment. Thus, owing to the lack of face-to-face transactions and returns, in some cases, customer protection unions or other related authorities do not necessarily have the ability to impose any legal restrictions (e.g., full returns). In addition, in the notion of product unconformity, the interpretation "merchandise did not match description" (Moorthy and Srinivasan 1995) is originally listed as a primary reason for returns in a consumer reports survey. In such a survey, the consumers' responses are by nature subjective, because these are the reasons that consumers perceive, not the reasons that consumers can report and credibly testify in court. Thus, in line with Moorthy and Srinivasan (1995), we do not impose any returns restriction on the product unconformity scenario.

Table 6 The Three Notions of Quality Risk and Their Differences

	Product defect	Product misfit	Product unconformity
Possible return restriction	At least full return	None	None
Residual value	No	Yes	Yes
Unambiguous quality improvement	Possible	Impossible	Possible

7.2. Summary of Findings

In this article, we offer an alternative rationale for why various Internet sellers may adopt different returns policies. We show that the manufacturer never finds it optimal to induce the low-segment consumers to return the product that functions well, even if the manufacturer is entitled to charge non-refundable fees or restocking fees; additionally, the manufacturer can design an appropriate refund so that products that function well never get back to the store. We also find that consumer returns are offered only when the high-segment consumers incur a higher hassle cost, the product quality is moderate, and the valuation of the low segment is moderate. Furthermore, when the product quality is relatively low, the manufacturer may over-compensate the consumers for returning the product; on the other hand, when the product quality is relatively high, further improvement in product quality may not induce a higher selling price. The restocking fee (measured in the percentage of selling price) may be non-monotonic in product quality, and the impacts of returns policies are ambiguous from the perspectives of either social welfare or consumer surplus. Finally, we investigate the alternative scenarios in which either the low-segment consumers are associated with a higher hassle cost or consumers may obtain some residual values from the inappropriate products.

7.3. Implications of Return Restrictions

In certain situations, the manufacturer may be bound by some legislation or physical restrictions on offering the returns policies (e.g., full returns restriction). In such a scenario, it is conceivable that the screening instrument is less effective, because it sometimes restricts the manufacturer's ability to appropriately adjust the price-refund bundle to differentiate consumers. More specifically, in section 4, we show that returns are useful for this screening purpose only when the product quality and the consumer type differential are both moderate. In the presence of return restrictions, the same result holds because the other two strategies BN and H are not affected, but strategy BR should be less profitable. In Figure 2, this implies that the region for strategy BR (using returns to differentiate consumers) should shrink. However, it

does not affect our qualitative findings—returns serve as a screening device only when the product quality and the consumer type differential are both moderate.

We can also articulate how this full return restriction factors into the quality choice in section 5. In Tables 4 and 5, we list three regions that represent different degrees of this screening effect. Specifically, the screening of returns is the least effective in Region III, which should be related to the case with full return restriction. This is because when the screening is less effective, the manufacturer's quality choice should put more weight on the other two strategies (BN and H), and these two strategies are unaffected by the full return restriction. As in Tables 4 and 5, we provide a comprehensive study of comparative statics, we expect that the results in Region III should answer directly to the reviewer's question on the full return restriction. As Region III arises when the screening is less effective, all the findings in that region can be directly related to those under some restrictions that deter the manufacturer from using returns to differentiate consumers.

7.4. Possible Extensions

Our analysis can be extended in a number of ways. For example, in our model, we focus exclusively on the monopoly setting. However, the reduced transaction costs and the prevalence of e-business easily invite competition that might be more intense than the traditional supply chains. Thus, it would be intriguing to investigate how the nature of competition affects the allocation as well as the return policy. In this scenario, consumers may exhibit store preferences and therefore a horizontally differentiated model may be more appropriate to describe the consumers' tastes. Second, in numerous supply chains, the products are sold through independent retailers. In such a scenario, an important strategic decision is whether the manufacturer or the retailer should accept the returns, and how to coordinate the supply chain in the presence of demand and valuation uncertainties. Finally, there are many other aspects of returns that we leave out in our analysis, for example, the time windows, exchange vs. returns, whether consumers are required to obtain Return Goods Authorization via email notification. Such an extension remains a research priority.

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Note

[1]The parameters we use to generate this figure are the following: $v_2 = 10$, $q = 0.5$, $T = 8$, $t = 3$, and $s = 1$; the results are robust to the specific choice of parameter combinations. Note that we have excluded the cost of reducing the quality risk, as this endogenous decision is investigated in the next section. If instead the cost of improving the quality risk is included, the region for strategy H will become larger because in this case, the manufacturer only bears the cost for serving the high segment, whereas in the other two cases, the cost is incurred for the entire market.

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